AN INSIGHT INTO CLOUD COMPUTING AND ITS MODELS

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***Abstract-* Cloud Computing is one of the trending topics in today's world. It brings about a major change on how the information can be stored and retrieved. Mega Corps such as Amazon, Google, and Microsoft also use clouds for smooth functioning. This paper covers the aspects of cloud computing and its models, benefits, and challenges faced while adopting cloud computing. It has undoubtedly taken commercial world by storm.**

Keywords- Cloud Computing, models, Amazon, Google, Microsoft.

1. INTRODUCTION

In simple words, cloud computing is nothing but fulfilling the user's demands for an application via Internet. Several applications of cloud computing include storage of information on the cloud and this information can be accessed by the user at any moment of time with an access to the Internet. All these operations can be carried out without the aid of our own computers [1].

Main highlight is that the end-user needn't possess the required technologies. Applications including email, wiki applications, and online tax calculation, utilize cloud computing. Cloud computing increases the efficiency of work done. Unlike the architecture of system models, cloud computing allows us to dynamically scale up or down depending on the scenario thereby offering increased reliability, faster response time and easier methods to handle internet traffic conditions [2]. It also supports multi-tenancy by offering systems that can be shared by several organizations or individuals.

Dependency is created by cloud computing simultaneously. These services are built in such a way where the relationship between technology and end-users is taken into consideration. End users must depend on the service provider to make sure that information is stored securely.

Fig.1 Basics of Cloud Computing

Cloud computing services mainly focus on users with meagre resources with an increasing requirement for information services.

2. CLOUD COMPUTING MODELS

Three types of cloud models shown in the fig.2 are:

1. SAAS
2. PAAS
3. IAAS

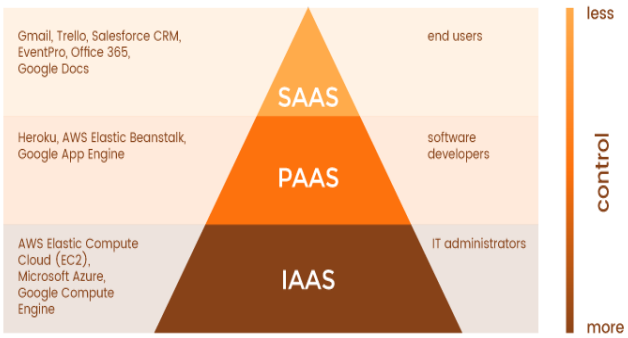

Fig2: Cloud computing models

2.1 SaaS (Software as a service)

Applications are distributed among the users as an app through the Internet. This is generally known as Software as a Service. SaaS users need not install a separate software to store, maintain or

update information as it will be readily available in the cloud [3]. The only prerequisite is a stable and fast Internet connection. Some well-known SaaS applications are Microsoft Office 365, Google Apps.

Some of the advantages SaaS are:

- Reduced cost – The users should subscribe to SaaS services to use it. But the services are free of license fee and hence saves money. Other hardware and software maintenance charges are reduced because they are managed by the administrator.
- Easy installation - Establishment of the SaaS app is done in the cloud. So, it saves time.

Few disadvantages of SaaS are:

- Control Issues – SaaS apps are hosted by a third-party source, this limits the amount of control that developers will have once the service is launched. The lack of this control causes issues for users, as they will be forced to keep up with updates and other system requirements.
- Security - The privacy of sensitive data is a major consideration in the cloud-hosted services.

2.2 PaaS (Platform as a Service)

In PaaS, the customer will have the freedom to build his own apps that can run on the service provider's infrastructure [4]. This Service provides a pre-defined OS and App Server to get the maximum capacity of applications.
Examples: Linux, Apache, MySQL, PHP, J2EE, Ruby etc. [3]

Advantages of PaaS are:

- The overall cost is low as the resources are allocated only on demand and servers are updated automatically.
- Applications are extremely secure as the systems are updated automatically.

Some of the disadvantages of PaaS are:

- There may be a difference in the environment among different cloud service providers, so the app should be adapt according to that.

2.3 IaaS (Infrastructure as a Service)

IaaS provides many computing resources in the form of network, storage, OS, hardware devices on demand. Customers of IaaS can use these services using internet [3].

Advantages of IaaS are stated below:

- The users can have access to the complete control of their application in this service. The best examples are Virtual Machines which can be completely adapted to the user's necessity.
- Highly efficient design and time portability is offered by the IaaS. Meaning, without and additional work an application can be transferred to Microsoft Azure.

Disadvantages of IaaS are:

- Since complete control is given to users, they show more interest to stick to particular version for the dependencies of applications. The migration of application to future versions would be difficult for the users.

3. BENEFITS

Cloud computing provides numerous benefits depicted in fig.3 is which are explained below.

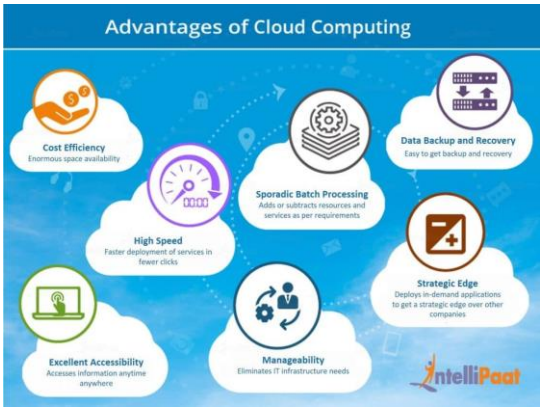

Fig.3 Benefits of Cloud Computing

3.1 Scalability

Generally, the user has to purchase and install expensive upgrades. But the cloud service provider does a favor to the user by handling this. This also frees up time for the user to do his works.

3.2 Collaboration

Cloud-computing plays a huge role in organizations having a wide range of employees where collaboration is the top priority. Cloud computing makes collaboration a simple process. Cloud computing allows multiple users to work simultaneously on the same data. All the members of the team can view and share information at the same time easily and more securely in case of a

cloud-based platform.

3.3 Security

Cloud-computing gives great security when any confidential data is lost. Cloud computing makes it possible to access the data using the Internet since it's saved in the cloud. Data which isn't required can be remotely wiped out safely.

3.4 Mobility

Information is stored in cloud spaces but not on a single computer. Hence, the information can be accessed and shared easily by the users at any point of time, from any device, and from anywhere.

4. LIMITATIONS

Though cloud computing has several advantages, it does have its own limitations. Some of them are discussed below.

Fig.4 Limitations of Cloud Computing

4.1. Risk of data confidentiality

In cloud computing, user's data is stored on servers where it can be operated by others **[5]** .This poses a threat on the cloud users. Hence confidentiality of data is treated as a serious issue.

4.2. Internet Connection Dependency

The most apparent drawback is that it is completely dependent on the network. This causes discontinuity in work. Similarly, a lot of bandwidth is used up by web-based apps, a low speed internet connection makes cloud computing worrying and often intolerable. Cloud computing is not suitable for slow internet connection.

4.3. Vulnerable in the Event of an Attack

Potential vulnerabilities are exposed in every cloud computing component. Many organizations suffer severe attacks and security threats from time to time.

4.4. Technical problem

Though cloud computing services are easy to use, there are problems that could be encountered due to technical issues. These issues can be solved only with the aid of a technician who wouldn't be available 24/7.

5. CONCLUSION

The applications of cloud computing are increasing day by day. As cloud computing industry continues to grow, this paper will provides an insight into cloud computing and its models as a whole there by paving the path for future work. This paper mainly focuses on the primary details of cloud computing. It starts off with an introduction to cloud computing followed by its models, and the corresponding merits and demerits are also discussed. In the present day, approximately all small and big industries including Google, Yahoo, Amazon, manage storage, traffic, and hardware requirements by deploying cloud computing. Thereby, the future of cloud computing is indeed bright.

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